

Three weeks of discussion have moved my thinking toward a claim the initial post approached but did not quite reach. Karyna established that unsupervised methods such as k-means carry their own normative assumptions through parameter selection, so neither paradigm offers an epistemically neutral starting point. Fatma's contribution sharpened this: CRISP-DM's linearisation denies critical review at the deployment stage. Accountability entered through Lana's response; EU non-discrimination frameworks, she noted, lack the procedural tools to handle harms distributed across data, design, and deployment contexts (Wachter, Mittelstadt and Russell, 2021). Ali pushed back most sharply. Even a k-means-first exploratory workflow, he argued, operates on unrepresentative data; bias is a collection problem that neither Random Forest nor k-means can resolve through architectural choice alone.

Ali's objection reoriented the argument. Celi et al. (2022) demonstrate that more than half of clinical AI datasets originate from the US or China, meaning representativeness is embedded before any practitioner makes a paradigm choice. Jones et al. (2023) extend this by showing that bias mitigation routinely fails because it addresses the model rather than the causal roots of data disparity. Schröer et al. (2021) supply the structural mechanism: deployment monitoring is absent in most CRISP-DM implementations. The stage at which bias surface is the one practitioners most consistently skip, consistent with the evaluation logic emphasised across Units 6–8. Mehrabi et al. (2021) confirm that measurement and historical bias (and, implicitly, label bias) are each embedded in data before a model is ever trained.

What the discussion surfaces is the structural limit of governance at the training or deployment stage: audit protocols detect deviation from intended behaviour but cannot detect exact conformity to a misspecified intention. The more uncomfortable implication, which none of the posts confronted, is that responsible deployment requires treating problem specification as an open negotiation rather than a technical precondition. The question is not which algorithm to deploy, but toward what outcome, defined by whom, and revisable under what conditions, though operationalising this institutionally remains unresolved.

References

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